

Computer Security - CheatSheet

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CompSec Properties - Confidentiality. prevent unauthorized disclosure of information. *authorized users may read a file*

- **Integrity.** prevent unauthorized modification of information. *authorized programs may write a file*

- **Availability.** prevent unauthorized denial of service or access to information and resources. *authorized services can access a file*

- **Authenticity.** assurance that entities (users, systems, or data) are genuine and can be verified as such.

- **Anonymity.** protection of an individual's identity from being disclosed or linked to specific actions or data.

- **Non-repudiation.** assurance that a party in a communication cannot deny the authenticity of their signature or the sending of a message.

The Adversary. malicious entity aiming at breaching the security policy and **will** choose the optimal way to use its ressources to mount an attack that violates the security properties.

Threat Model. describes the ressources available to the adversary and their capabilities *(has access to internet, but doesn't have access to the internal network of the company.)*

Threat. Who might attack which assets, using what resources, with what goal, how, and with what probability

Vulnerability. Specific weakness that adversaries could exploit with interest in a lot of assets *(API is not protected, password appears in plain text...)*

Principles of CompSec.

- 1. **Economy of mechanism** Keep security mechanism design simple and small ⇒ Easier to audit and verify, testing is **not** appropriate to evaluate security.
- **Trusted Computing Base (TCB).** Every component of the system on which the security policy relies upon *hardware, firmware, software.*
The TCB is trusted to operate correctly for the security policy to hold. → If something goes wrong in it, the security policy may be violated
- must** be kept small to easy verification / diminish the attack surface
- 2. **Fail-safe defaults.** Base access decisions on permission rather than exclusion. *(Whitelist, do not blacklist)*
If something fails, be as secure as it does not fail errors / uncertainty should error on the side of the security policy.
- 3. **Complete mediation..** Every access to every object must be checked for authority. A **Reference Monitor** mediates all actions from subjects on objects and ensures they are according to the policy. *△time to check vs. time to use*
- 4. **Open design** The design should not be secret *Always design as if the enemy knows the system. (Crypto-only secret is key.Authentication-Only secret is password.for Obfuscation-Only used secret is noise.).assuming the thread model can't get a hold of the system is unrealistic (employee corruption, ...)*

Access Control. Security mechanism that ensures all accesses and actions on objects by principals are within the security policy.

no chicken soup *(checks everywhere in code), use a reference monitor, module used all over code checking for subjects and actions*

Discretionary Access Control (DAC). Object owners assign permissions, ownership of resources. *Linux, Social Networks*

Mandatory Access Control (MAC). Central security policy assigns permissions, usually for organizations with need for central control. *Military, Hospital, ...*

Access Control Matrix. abstract representation of all permitted triplets of (subject, object, access right) within a system.

	file1	file2	file3
Alice	read, write		read
Bob		read, write	read, write

Complexity. O(f · u)

Access Control List (ACL). associate permissions to objects, stores permissions close to the resource.

file1: (Alice,read/write), file2: (Bob, read/write), file3: (Alice,read),(Bob,read/write)

⊗ easy to determine who can access a resource and to revoke rights by resource.

⊗ hard to check all users rights, to remove all perms. for a user, delegate perms.

Role Based Access Control (RBAC). access granted based on user roles and predefined permissions. systems have too many subjects (that come and go) → large dynamic ACLs. Subjects are often similar to each other and get assigned the same rights.

- 1. Assign permissions to roles,
- 2. Assign roles to subjects

3. Subjects have the permissions of their assigned role

⊗ role explosion (temptation to create fine grained roles), limited expressiveness, difficult to implement separation of privilege + least privilege.

Group Based Access Control (GBAC). access granted based on user group membership. **exactly like RBAC but instead of roles, permissions are assigned to groups.** groups are broader, represent organizational units instead than specific functions.

- 1. Assign permissions to groups
- 2. Assign subjects to groups

3. Subjects have the combined permissions of all their groups

⊗ coarse granularity, overlapping group memberships, inconsistent permissions, difficult to manage if users belong to many groups.

*In case of **Negative Permissions** check negative permissions first before group, △system crashes before negative checks.*

Ambient Authority. an action succeeds if the subject only specifies the *operation* and the *object name*, not the specific authority used.

⊗ leads to accidental misuse of authority, programs may act with more rights than intended. → Confused Deputy Problem.

Confused Deputy Problem. when a program (deputy) is tricked into misusing its authority on behalf of another subject.

Alice runs compiler(input, bill) ⇒ compiler (with write access to bill) overwrites billing file. ⇒ Alice uses compiler's authority to modify bill indirectly.

⊗ ambient authority allows unintended privilege use.

Solutions:

- 1. Restrict privileged process access.
- 2. Make privileged process check user's authorization.
- 3. Use **Capabilities** to explicitly delegate rights.

Harm. The bad thing that could happen when the **threat** materialsizes. *(adversary steals the money, learns my password...)*

Security Policy. high level description of the security properties that must hold in the system in relation to assets and principals.

- **Assets (objects).** anything with value (data, files, memory) needing protection.

- **Principals (subjects).** people, computer programs, services.

Security Mechanism. Technical mechanism used to ensure that the security policy is not violated by an adversary within the threat model, **we can only prepare for threats we're aware of**

(Policy. ensure messages cannot be read by anyone but the sender and the receiver, Mechanism. encrypt the message before sending)

Composition of Security Mechanisms

- **Defence in depth.** As long as one remains unbroken the Security Policy isn't broken) *(two-factor auth)*
- **Weakest Link.** if anyone fails the Security Policy, it is broken. *(security questions for a lost password → just need to know the answer...)*

Humans can be vulnerabilities - phishing attacks, bad use of passwords...)

To show a system is secure. (under a specific threat model)

- Attacker - Just one way to violate **one** security property is enough.
- Defender - No adversary strategy can violate the security policy.

Security Argument. Rigorous argument that security mechanisms in place are effective in maintaining security policy subject to assumptions on Threat Model.

- 5. **Separation of privilege.** No single accident, deception, or breach of trust is sufficient to compromise the protected information
- **Privilege.** A privilege allows a user to perform an action on a computer system that may have security consequences. *(create a file in a directory...)*
- 6. **Least Privilege.** Every program and every user of the system should operate using the least set of privileges necessary to complete the job. *Rights are added on need, discarded after use. Users should only know about things if they have to.*
- 7. **Least Common Mechanism** Minimize the amount of mechanism common to more than one user and depended on by all users. Every shared mechanism represents a potential information path between users.
- 8. **Psychological acceptability.** It is essential that the human interface be designed for ease of use, so that users routinely and automatically apply the protection mechanisms correctly. *(hide complexity, cultural acceptability...)*
- 9. **Work Factor**
Compare the cost of breaking the mechanism with the resources of a potential attacker. *(cost of compromising insiders, cost of finding a bug, monetization...)*
- 10. **Compromise recording** Detect and record security breaches with tamper-evident logs, ensuring traceability, integrity, confidentiality, and availability of recorded data.

User & Group Identities. Most modern systems rely on DAC.

UIDs / GIDs. numerical identifiers for users and groups.

/etc/passwd: username:password:UID:GID:info:home:shell

/etc/group: defines secondary groups.

Each user has a home directory and belongs to one or more groups.

UNIX Model. Everything is a file.

directories		owner	group	others	owner	group					
files	drwxrwxr-x	1	catronco	catronco	4096	Sep 16 14:23	exampledir				
	-rwxrwxr-w-	1	catronco	catronco	8600	Sep 15 15:20	hello				
	-rw-rw-rw-	1	catronco	catronco	150	Sep 15 15:14	hello.c				
	-rw-rw--w-	1	catronco	catronco	45	Sep 15 15:07	test1.txt				
		links		size		last modified	filename				

Directories. Read → list files; Write → add/remove files; Exec → traverse (cd).

Permission check order: 1. If process UID = file owner → check owner bits.

2. Else if GID matches → check group bits.

3. Else → check “other” bits.

Root (UID 0. bypasses all checks.

Changing Permissions.

chmod. modify permission bits. *chmod +r file, chmod 666 file.*

chown. change file owner/group(opt.) *chown root:staff /srv/config*

chgrp. change file group. *chgrp www-data /var/www*

Special Rights.

suid (set user ID). run program with owner's privileges. *puts s in owner's x field*

chmod u+s filename

allows normal users to change passwords without full root access.

sgid (set group ID). run program with group's privileges.

on a directory - creating a file inside gets folder's group

d rwx r-s r-x 2 root staff /srv/shared

sticky bit. on directories, prevents deleting/renameing files you don't own.

d rwx rwx rwt 10 root tmp /tmp

Special Users.

root: UID 0, full privileges, bypasses checks, in TCB.

sudo run as root *su [username]* switch to account(default is root)

nobody: UID -2, owns no files, minimal privileges, safer for untrusted code.(least privilege)

⊗ flexible, simple model, widely adopted.

⊗ relies on **ambient authority**, prone to Confused Deputy attacks.

Windows: DACL. Controls access to objects via list of ACEs.

ACE (Access Control Entry). <Type, Principal, Permissions, Flags>.

- **Types.** Allow / Deny (negative / positive). Deny takes precedence and ordered with Denied permissions first.
- **Principal.** User or group (SID).
- **Permissions.** Fine-grained rights(*Read, Write, Execute, Delete...*)
- **Flags.** Inheritance, propagation, object-specific.

Access Tokens. Thread/process carries user + group SIDs checked against DACL.

Least Privilege. Users run limited by default; elevation via “Run as admin.”

Capabilities. associate permissions to *subjects*

Alice: {(file1, read/write)} Bob: {(file2, read/write), (file3, read/write)}

⊗ easy to determine all permissions of a user and to delegate rights by subject.

⊗ hard to determine who can access a resource or to revoke rights by resource.

Security Model. a design pattern for a set of properties. (△ not covered by model - who are the subjects? what are the objects? what mechanisms to use to implement it?)

Bell-La Padula (BLP).
security model where Subjects S and objects O are associated to a level of **Confidentiality**.
Access rights. Execute, Read, Append, Write.
- Objects are associated to a Security Level = (Classification, set of categories).
{{Unclassified < Confidentiality < Secret < Top Secret}, {NATO, Crypto, Nuclear}}
Dominance Relationship.
Transitive, There always is a Top and Bottom, Only partial ordering (some pairs of elements can't be compared).
A security level (l_1, c_1 . dominates (l_2, c_2 . if and only if $l_2 \leq l_1$ and $c_2 \leq c_1$.
eg. {Secret, {Nuclear, Army}} ≥ (Confidential, {Army})
Clearance. max security level a subject has been assigned. *clearance-level(S_i)*.
Current Security level. subjects can operate at lower security levels. *current-level(S_i)*.
Declassification. Removing classification labels from information so it can be shared with lower security levels. *soldier (S) with Secret clearance shares sanitized info like "We've seen tanks moving" with a lower-level user (U)*.

Covert Channels. Any channel that allows information flows contrary to the security policy, *Storage channels(shared counters, ...), Timing channels(queueing time. ...)*.
△Least Common Mechanism. *mitigated by adding noise or isolation(no high → low level communication)*.

BIBA (Integrity).
Security model where subjects S and objects O are 1. associated to a level of **Integrity** (trustworthiness).
Access rights. Read (Observe), Write (Modify)
BIBA to create Integrity Policies.
Simple Integrity Property.
If (s, o, r) is a current access, then level(s) dominates level(o).
(**SUBJECTS CAN'T READ DOWN**) ⇒ prevents contamination from low to high.
***-Integrity Property.**
If (s, o, w) is a current access, then level(o) *does not* dominate level(s).
(**SUBJECTS CAN'T WRITE UP**) ⇒ prevents low from corrupting high state.
Discretionary Property (DAC). Subject must still have explicit permission for the access (r, w, x) per the access control matrix.
Basic Integrity Theorem (induction). If the initial state is integrity-secure and all state transitions satisfy BIBA properties, then all reachable states preserve integrity.

Sanitization (lifting low → high).
Fail-safe default: deny by default; elevate only after checks pass.
- *Whitelist over blacklist:* validate that all required properties of "good" hold.
- *Context-aware:* encoding, schema, range, semantics, and provenance checks.
Note: Sanitization bugs commonly break integrity guarantees.
Principles for Integrity.
- **Separation of Duties.** Require multiple principals to perform an operation
- **Rotation of Duties.** Allow a principal only a limited time on any particular role and limit other actions while in this role
- **Secure Logging.** Tamper evident log to recover from integrity failures. Consistency of log across multiple entities is key.

Chinese Wall Model. All objects are associated with a label denoting their origin.
(Pepsi, Coca-Cola, Microsoft Audit, Microsoft Investments)
Define **conflict sets** of labels. *{Pepsi, Coca-Cola}, {Microsoft Audit, Microsoft Investments}*.
Subjects are associated with a **history** of their accesses to objects and their labels.
Access Rules. A subject can access an object (read or write) **only if** the access does not allow information flow between items with labels in the same conflict set.

Example (Direct Flow).
1. Access to Pepsi (OK)
2. Access to Microsoft Invest (OK)
3. Access to Coca-Cola (Denied)
Example (Indirect Flow).
1. Alice accesses Pepsi
2. Bob accesses Coca-Cola and IBM
3. Alice tries to access IBM (*Denied, indirect link via Bob*)

Sanitization.
Allows more flexibility by "un-labeling" some items→controlled sharing/reuse of data.

All together (Asymmetric Cryptography) - Bob sends a message to Alice
1. Get public keys (via PKI).
Bob retrieves Alice's encryption public key PK_{Alice} and verification public key PK_{Alice} .
2. Prepare the message.
Bob has message M and computes its hash $h(M)$.
Allows: the signature to cover the exact content of M while keeping computation efficient.
3. Sign the message hash.
Bob uses his secret signing key SK_{Bob} to generate a digital signature. $\text{Sign}_{SK_{\text{Bob}}}(h(M))$.
4. Encrypt the message.
Bob encrypts the message using Alice's encryption public key PK_{Alice} : $E_{PK_{\text{Alice}}}(M)$.
Allows: only Alice, who holds the matching secret key SK_{Alice} , to read the message.
Authenticity. only sender can generate a valid signature, allowing the server to verify the sender's identity and preventing impersonation. **Integrity.** Any message change invalidates signature, ensuring tampering is detectable. **Non-repudiation.** only sender can produce the valid signature → they can't deny later having sent the message.

Block Cipher - Electronic Code Block (ECB).
encrypt each message block independently using the same key.
1. The message M is divided into blocks.
 $M = M_1 M_2 M_3 M_4$.
2. Each block is encrypted separately $C_i = E_K(M_i)$.
3. Decryption. $M_i = D_K(C_i)$.
△If ($M_i = M_j$) ⇒ ($C_i = C_j$) ⇒ (freq. analysis)

BLP to create Confidentiality Policies.
Simple Security Property.
if (subject, object, w/r) is a current access, ⇒ level(subject) dominates level(object).
(**SUBJECTS CAN'T READ UP**)
Star Property. If a subject has simultaneous "observe" (r,w) access O_1 and "alter" (a, w) access to O_2 then level O_2 dominates level O_1 .
(**SUBJECTS CAN'T WRITE DOWN**). *changing object's perms is write-like*.

Discretionary Property. A subject may only access an object if it has explicit permission for that specific type of access (r, w, x, etc.) defined by the access control matrix.

Basic Security Theorem(induction). if all state transitions are secure, and the initial state is secure, then every subsequent state is secure regardless of the input.

BIBA Low-Water-Mark Variants (taint-tracking).
For Subjects.
Subjects start at their max integrity, **on read:** $\text{current-level}(s) = \min(\text{current-level}(s), \text{level}(o))$.
⇒ once tainted, subject sinks; prevents writing up thereafter.
For Objects. On write by s:
 $\text{level}(o) = \min(\text{level}(o), \text{level}(s))$.
⇒ objects "sink" easily; can cause integrity collapse.
Mitigation: replicate high/low objects; sanitize before promotion; detect/flag unexpected level drops.

BIBA Additional Actions - Invoke.
Simple Invocation. only allow subjects to invoke subjects with a label they dominate.
⊕ protect high integrity data from misues by low integrity principals
⊖ what level is the output ?
Controlled Invocation. Only allow subjects to invoke subjects that dominate them.
⊕ prevents corruption of high integrity data
⊖ hard to detect polluting information.

Cryptography.
Data in transit. Securing communications. **Data at rest.** Securing stored informations
Let C the Ciphertext, K the Key, M the Message/Plaintext. $C = E_K(M)$ and $M = D_K(C)$
Keyspace. Number of possible keys that can be used in an encryption algorithm.
Invertibility Requirement. $\forall K, M | D_K(E_K(M)) = M$ (otherwise can't recover message)
Security Requirement. Functions should be hard to invert without knowing K.
Ideal Case - adversary must try every possible combination of keys (bruteforce).
Caesar's Cipher
- Encryption. Shift each letter by a fixed number (K)
- Decryption. Shift each letter by a fixed number (-K)
Keyspace. Only 25 possible keys.
 $\log_2(25) = 4.6$ bits of security (too small).
Both can be broken using **Frequency Analysis Attack**.

Frequency Analysis. Use statistical properties of the language.
1. Most frequent letter in English is 'e'. 2. Identify most frequent letter in Ciphertext. 3. Map it to 'e'.
Ideally, An N-bit key should offer security as close to N bits as possible (require 2^N attempts), if not the algorithm is considered **broken**.
Types of Adversaries. security models
- Passive Eavesdropper - adversary can only read the ciphertext
- Active Attacker - adversary can influence the system (*corruption of one of parties, ...*)
Known Plaintext Attack (KPA). **Chosen Plaintext Attack (CPA).**
Active security model. Attacker gets - Suppose Eve convinces Alice to encrypt chosen messages access to some pairs, (message, cipher- with the secret key text), all encrypted with secret K.
- For all m , Eve gets access to $E_K(m)$. Eve has access to an **En-F**rom these pairs, she tries to guess key crypton Oracle.
K to decrypt other messages (*realistic, she could get access to an encrypted messaging app realistic because of message headers, for limited time, or to an encryption api*).
message headers ("From:...", times-△if Eve encrypts entire alphabet she can reveal the key instantly).
Side-Channels Attacks
- Timing attack - Eve measure how long it takes for a given message to get encrypted or a ciphertext to be decrypted
- Power Analysis - Eve observes the energy consumed by the device doing the crypto.

One-Time-Pad (OTP). OTP is **Perfectly Secure**.
Goal - Remove frequency analysis
- Use a key of random bits as along as the message is equally likely
sage.
 $\text{Enc}(k, m) = m \oplus k$
 $\text{Dec}(k, m) = \text{Enc}(k, m) \oplus k$
Key should be random for every sent message $\forall m, c, P(M = m | E_k(m) = c) = P(M = m)$.
sage. Otherwise attacker can collect information- **Guarantees confidentiality** about the message.
Integrity Attack Example (Eve flips the first bit of M):
1. Eve flips the first of C to get C'
2. Bob decrypts C': $M' = C' \oplus K = (M \oplus K \oplus \Delta) \oplus K = M \oplus \Delta$
Symmetric Cryptography Schemes **Symmetric Cryptographic key**
Encryption/decryption done with the **same key**
- **Block Ciphers.** Operate on fixed-size blocks.
- **Stream Ciphers.** Operate on bit/byte at a time, like pseudo-OTP.
- **Known to both parties.** Partners must agree on the key before starting using the primitive
- **Reused.** The keys is pre-shared once and then reused (*but keys have a "duration"*)
- **Must be secret.** Revealing the key eliminates any protection provided by the primitive

Stream Ciphers - The pseudo-OTP. emulate OTP while solving key-length problem.
1. Secret short Key (K) shared between Alice and Bob 2. Key Stream Generator - Uses K and Initialization Vector (IV) → an arbitrary long, pseudo-random bit stream (S). 3. Encryption. $C = M \oplus S$ (*generator needs a key as main seed to generate a predictable random sequence, an iv for the sequence to start differently on each run*)
⊕ Speed, Low Error propagation(errors in one bit do not affect subsequent symbols)
⊖ Low Diffusion (a change in a bit only affects one bit), Susceptibility to Modification (low diffusion makes it easier to tamper)
⊖ key stream generators are periodic because seed is finite. Thus, we need period long enough to not an issue (avoiding frequency analysis)

Linear Feedback Shift Register for Key Stream Generators. build a ""random"" sequence of bits using a linear recurrence relation on a sequence of bit.
Example: Starting with state $a_0, a_1, a_2, a_3 | a_n = a_{n-3} \oplus a_{n-4}$ (⊕ Easy to build/analyze.)
Randomness of LFSR. **Distribution property.**
if characteristic polynomial of the recurrence relation of the LFSR is primitive. The maximum possible number of states before repeating. For an L-bit register, the maximum period is $2^L - 1$ states
As a consequence, the sequence generated exhibit good distribution properties. Every possible non-zero state appears exactly once in a cycle.
⊖ underlying operation is **linear**. (*some algorithms can recover this with a stream subsequence*)

Symmetric ciphers are fast, but require a **pre-shared secret key**
Diffie-Hellman Setup. Alice, Bob (and Eve) know large public parameters: prime p/generator g
Alice's Actions
1. Chooses **private** secret a.
2. Computes Public Value $A = g^a \text{ mod } p$
3. Sends A to Bob.
Eve sees A, B, g, and p, but can't find a/b due to the *Discrete Log Problem*.
Both Alice and Bob can compute a shared secret K: $K = B^a \text{ (mod } p) = (g^b)^a \text{ mod } p = A^b \text{ (mod } p)$ this is uses **Trapdoor functions**, easy to compute but hard to revert.

Man-In-The-Middle (MITM). 1. Eve sends E_B to Alice. Alice computes K_{AE}
Vanilla DH is Vulnerable to Active Attack, 2. Eve sends E_A . Bob computes K_{BE}
DH guarantees key agreement, **not** authenticity. 3. Creates an Alice-Eve channel and Eve-Bob channel
Attacker could intercept A from Alice and B she can read, modify, and relay all communication. from Bob, and then impersonate.

Asymmetric Cryptography. Encryption using a public-private key pair. Public key can be stored on a public server, the **Public Key Infrastructure (PKI)**.
Hash Functions. maps any-length message to a fixed-size output using a hash function. $h(M) = H$
Security properties. 1. **Pre-image resistance.** Given $H = h(M)$, it is hard to find M.
2. **Second pre-image resistance.** Given M, it is hard to find another $M' \neq M$ such that $h(M') = h(M)$.
3. **Collision resistance.** It is hard to find any two values M and M' such that $h(M) = h(M')$.

Block Cipher - Counter Mode **Block Cipher - CBC-MAC.** turns block cipher into MAC by chaining add randomness per-block without using message blocks with encryption.
inter-block dependencies. $C_0 = IV$ **Encrypt.** $C_i = E_K(M_i \oplus C_{i-1})$ MAC $_K(M_1, \dots, M_n) = C_n$ (parallel) deterministic. fixed IV → same message/same MAC.
Encrypt. $C_i = M_i \oplus E_K(\text{Nonce} + i)$. △message length must be known and fixed; otherwise, extension **Decrypt.** $M_i = C_i \oplus E_K(\text{Nonce} + i)$. attacks are possible (e.g., using $M_i(T \oplus M')$).
Nonce must be unique (never with same key) reuse leaks keystream.

Message Authentication Code **Code** **hertext.** 2. **MAC-then-Encrypt** send $E_K(P_1 P_2 || MAC_K(P_1 P_2))$. (MAC). short tag computed from hides MAC and provides confidentiality/integrity on plaintext but a message and secret key(pre-no integrity ciphertext shared) to verify message **integrity** 3. **Encrypt-then-MAC** send $E_K(P_1 P_2) || MAC_K(E_K(P_1 P_2))$. **authenticates** ciphertext, ensures **confidentiality/integrity** ✓

Authentication. the process of verifying a claimed identity. (Prove who you are.)
Passwords. secret shared between user and system.
Problems:
I **Secure transfer.** password may be eavesdropped when communicated.
Encrypting makes password unreadable to eavesdropper, but still allows copy and send. (△**Replay-Attacks.** sending same request without needing the password)
Challenge-Response protocol solves this.
1. Principal declares they want to login.
2. System authenticating principal sends a **Challenge** (a random value).
3. Principal encrypts password *together* with the challenge.(ensures password is never encrypted with same value and encryption is fresh)
4. After checking password, System *must* delete the challenge.
II **Secure check.** naïve checks may leak information about the password.
Check Options:
1. Letter by letter - Time to check leaks the number of correct checked letters. (Timing channel)
2. Check everything - Same time regardless of input (hashing solves this by default).
III **Secure storage.** if stolen entire system is compromised.
Password must be stored to be checked against but *not stored in clear* → an adversary can read them and impersonate users.
Storage Options :
1. Encrypted Passwords. requires key to be stored in same system as database, not secure.
2. Hashed Passwords. Not a solution, hashing does **not** require a key. △passwords are not truly random.→Database can be downloaded and compared against hash of most common passwords (**Dictionary Attack**).
3. Hashed + Salted Passwords (Do this !). **Salt.** unique random value generated per password. **We store:** (username, $H(123456 || salt_n), salt_n)$.
Dictionary attacks are still possible but same passwords are undetectable.
Principal asks to login→System sends Challenge→Principal sends encrypted password with Challenge →System checks Challenge, decrypts, looks up hash by username, hashes plaintext with salt and checks if same as stored.
IV **Secure passwords.** easy-to-remember passwords tend to be easy to guess
Biometrics. measurement and statistical analysis of people's unique physical characteristics(fingerprint, face, ...).

Biometrics Authentication.
1. **Enrollment.** biometric is captured by a sensor and processed to be converted in a stream of bits(Biometric Template). **Capture → Process → Store.**
2. **Verification.** biometric is captured, processed and compared against the stored one. **Capture → Process → Compare.** (threshold for match vs. no match).
Balance between False Positives (wrongly authenticated) and False Negatives(wrongly rejected).
Storing / Processing can happen locally(⊕ privacy) / remotely(⊕ security).

Authentication Tokens. by proving ownership of a token. (Smart Card, ...)
1. Offline - Device and Server establish a common *seed* (RANDOM value) & synchronize their clocks.
2. When Device wants identity:
(a) Server asks to prove it. (b) They both compute $n = \frac{now-start}{interval}$. *Current interval.*
(c) Device applies **keyed** function (pre-shared key) n times on initial seed. $v = f^n(seed)$
(d) Device sends result to Server. (encrypted, eavesdropper can't have the seed).
(e) Server computes same value and checks against the one received.
Can't be a hashing function (hash functions aren't keyed!, eavesdropper could compute $h_k(seed)$ for any k after the one observed.).

Security Engineering process. **Attack Engineering process.**
1.Define security policy (*principals, assets, properties*) and exploit weaknesses introduced during the security engineering process due to lack of careness/knowledge or errors.
a threat model. 2.Define security mechanisms that support policy of threat model. 3.Build implementation to support the mechanisms
Threat Modeling. process to identify threats / unprotected resources.
- **Attack Trees.** represent goal as the root and ways to achieve it as branches.
- **STRIDE.** 1) Identify assets, principals, trust boundaries, and named flows 2) **For each threat:**
1. STRIDE category, 2.what changes/abuse happens, 3.which flow it affects, 4.concrete mitigation.
Spoofing: fake identity (user/service). **Tampering:** unauthorized edit of data/messages/code.
Repudiation: deny action (no provable record). **Info disclosure:** leak confidential data/metadata.
Denial of service: exhaust/block resources/reduce availability. **Elevation of privilege:** gain higher privileges than allowed.
- **PASTA.** start from buisness goals, processes, and use cases. Find threats in buisness model, assess impact, prioritize on risk.

Methods.
- GET. requests a resource; parameters are appended to the URL as query key–value pairs after "?", separated by '&'.
`http://www.mywebsite.com/apparel/skirt.php?sku=123&lang=en§=silk`
protocol hostname / domain directory filename query parameters
- **POST.** creates or updates a resource; sends data in the request body (not the URL) and typically modifies server data.
POST /test/demo_form.php HTTP/1.1
Host: w3schools.com
name1=value1&name2=value2
URL. standardized address that specifies a resource's access method, host, and path.
HTML. markup language using tags to structure content and tell browser how to render a webpage.
WEF. Wi-Fi security protocol that encrypts frames with **RC4** (stream cipher with random 24-bit IV). IV repeats frequently, causing keystream reuse and enabling message (and sometimes key) recovery.
PHP. server-side scripting language for dynamic web pages; uses variables and request/session data ("\$_gatewayET", "\$_POST", "\$_SESSION") and outputs HTML with 'echo'.
Lifetime of a GET/POST request.
1. Browser sends HTTP GET/POST request.
2. Web Server receives requests and identifies if it's a static asset(.jpg,...) or dynamic(.php).
3. PHP interpreter executes called .php 4. Sends response to web browser

Data Execution Prevention (DEP). Mark memory pages as writable OR executable, never both. (pre- vents code injection by making stack/heap non-executable)

⊗ low-overhead/hardware enforced
⊙ no self-modifying code.

Common Weakness Enumeration (CWE). database of software errors leading to vulnerabilities.
1. **CWE 1 - Insecure Interaction Between Components.** unchecked information sent between different components.
(a) **CWE 78 - OS Command Injection.** application embeds unvalidated user input into an OS command, allowing attackers to execute arbitrary commands *Is /home/user.*
(b) **CWE 79 - Cross Site Scripting.** application embeds unvalidated user input into dynamically generated web content, causes browser to execute injected scripts.
Useful Functions.
`navigator.userAgent`
Returns the browser identification string; increases fingerprinting surface.
`Chrome/143.0.0.0 Mobile Safari/537.36`
`window.location.replace(url)`
Forces navigation to url without adding a history entry.
`document.cookie`
Returns all non-HttpOnly cookies for the current origin; exposed session/state cookies
(c) **CWE 362 - Cross Site Request Forgery.** malicious website makes a state-changing action using user's session cookie for another website.
Same Origin Policy (SOP). web browser security mechanism that restricts scripts from one origin from reading data from another origin. An origin is a (protocol, host, port) *http:example.com.*
This doesn't prevent CSRF as it's based on sending a request and that the browser naturally attaches cookies of the destination domain in request (unless special setting).

Solutions:
For every state-changing endpoint:
- Check Origin header and only allow exact origin.
- If Origin missing, require Referer and ensure it starts with expected origin, else reject.
Keep GET side-effect free; Require a CSRF token (challenge) on every state-changing request; For high-riks actions require re-auth.
2. **CWE 2 - Risky Ressource Management.** not properly managed creation, usage, transfer, or destruction of important system resources
3. **CWE 3 - Porous Defenses.** defensive techniques misused, abused, or just ignored

Software Security.
Stack.Call frames: locals, args, return addresses
Heap.Dynamic allocations via malloc/new
Uninitialized data (BSS). Global/static zero-initialized storage; zeroed at program start.
Initialized data. Global/static with explicit initial values; loaded from the executable.
Text. Executable code (often read-only); mapped from the executable.
Args & env. argv/environment strings placed at startup near the initial stack.
Memory Corruption.
Unintended modification of memory location due to missing / faulty safety checks.

Memory Safety - Temporal Error. tried access to memory initially reserved for our program but not reserved anymore.

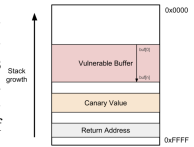
```
void vuln(int user1, int*array){  
  // no bound check for user1  
  array[user1] = 42;}  
  
void vulnerable(char* buf) {  
  free(buf); // variable was freed.  
  buf[12] = 42;}  
  
CWE 134 - Uncontrolled Format String.  
inject format specifiers that can leak memory contents in untrusted input string.  
"%4$p":Reads 4th parameter even if doesn't exist  
"%6$n":Write to address pointed to by 6th parameter  
  
int main(int argc, char** argv){  
  char buffer[100];  
  strcpy(buffer, argv[1]);  
  printf(buffer);  
  return 0;}
```

Code Injection Attack.
1. Control unchecked variable.
2.Overwrite return address. 3.Hijack control flow.
4. Arbitrary code execution.

```
void vuln(char* u1){  
  // strlen(u1) < MAX?  
  char tmp[MAX];  
  strcpy(tmp, u1);...  
} vuln(&exploit);
```


Attack Chain:
1. **Memory Safety Violation.** strcpy writes past tmp[MAX] (spatial error)
2. **Integrity (*C).** Attacker overwrites return address on stack
3. **Location (&C).** Return address now points to shellcode
4. **Usage (*&C).** CPU dereferences corrupted pointer
5. **Code Injection.** Shellcode executes with program privileges

Address Space Layout Randomization (ASLR). Loader/OS randomizes memory regions/code/stack/heap/libraries) at load time. Same code, different addresses on each run. (prevents target addresses prediction/breaks gadgets)
Limitations: 1.Probabilistic defense, not deterministic. 2.Vulnerable to addresses leaks. 3.Some regions remain static x86.
4.Performance overhead.(requires the system to keep track of all pointers)
Mitigation. A defense mechanism that prevents or reduces the impact of attacks.
1. **Effectiveness.** must prevent/defend attacks
2. **Efficiency:** low comp/memory overhead
3. **Compatibility:** easy deployment (no major software/hardware changes)
Examples: Sandboxing. isolates process from system resources and other processes.
DEP, ASLR, Stack canaries.
Stack Canaries. Compiler places random secret value between local buffers and return address. Checked before function returns and abort if corrupted.
(detects buffer overflows before control hijack)
Limitations: 1.Probabilistic, vulnerable to canary leaks. 2.No protection against targeted writes (jumps over canary). 3.No protection against reads. 4.Performance overhead from checks.



Error. difference between observed behavior and specified behavior (a violation of underlying spec)
Testing. executing a program to find bugs/errors and test security properties.
Ideally, complete test of all:

- **Control-flows.** all path of program.
- **Data-flow.** all values used at each location.

Practice is limited by *state explosion* (impossible to specify behavior)
Automated Testing. decided algorithmically (*run program/find bugs/enforce properties*).
- **Static analysis.** analyse program without executing it. (imprecise)
- **Symbolic analysis.(doesn't scale)** converts program to logical equations covering many executions, provides full path coverage.(complete). Path explosion: each branch doubles paths—exponential blowup.
- **Dynamic analysis.** *fuzzing* inspect program by executing it (hard to cover all paths)
Fuzzing. random testing that mutates input to improve test coverage (scales very well).
State-of-the-art fuzzers use coverage as feedback to guide input mutations.
Types of Fuzzing:
1.**Dumb Fuzzing:** not aware of input structure; randomly mutates input without guidance.
2.**Generation-based Fuzzing:** uses model describing valid inputs; generates new input seeds in each round based on model specifications.
Symbolic Execution with Fuzzing. Whitebox fuzzing uses *Symbolic Analysis* to generate new seed inputs for unexplored paths when random mutation gets stuck, improving coverage.

Sanitization. detect bugs by enforcing policies, **increasing testing effectiveness.**

- Assertions
- Segmentation faults
- Division by zero traps
- Uncaught exceptions
- Mitigations triggering termination

cannot detect logic bugs/specification, only bugs that violate specific policy

Network Security Properties.
-**Naming Security.** association between lower level names (*network addresses*) and higher level names (*Alice*) must not be influenced by adversary. (*Integrity, Authenticity, Availability(naming service)*)
-**Routing Security.** route over network and delivery of messages must not be influenced by adversary. (*Integrity, Authenticity, Availability, Authorization*)
-**Session Security.** messages within the same session cannot be modified (keep ordering and no adding/removing messages) (*Integrity, Authenticity*).
- **Content Security.** content of messages must not be readable or influenced by adversaries (*Confidentiality, Integrity, Authenticity*).

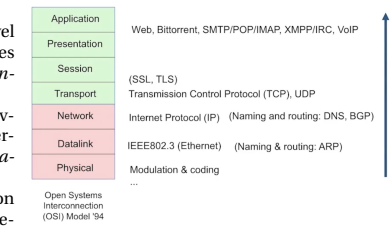
Subnet Mask. Used to determine network portion of IP address via bitwise AND operation.
Gateway. Router IP address used to forward packets to different subnets.
ARP (Address Resolution Protocol). translates IP addresses → MAC addresses. Each host maintains cache table of IP → MAC mappings. If not available, broadcast ARP request.
ARP Packet Format. (⚠️ No integrity checks nor authentication in ARP packets)
HTYPE (2 bytes): Hardware type (1 = Ethernet/Wi-Fi)
PTYPE (2 bytes): Protocol type (0x0800 = IPv4)
HLEN (1 byte): Hardware address length (6 for MAC)
PLEN (1 byte): Protocol address length (4 for IPv4)
OPERATION (2 bytes): 1 = Request, 2 = Reply
ARP Spoofing Attacks. (*Naïve threat model assuming insiders are trustworthy.*)
Man-in-the-Middle. Attacker convinces both parties their MAC is the other's MAC → intercept/modify all traffic
Denial of Service. Provide wrong MAC mappings → victims cannot receive packets
Resource Theft. Impersonate others to bypass bandwidth limits or steal allocated resources

IP Routing. Input:(IP_{source}, IP_{dest} , Subnet Mask)
1.Compute ($IP_{source} \wedge M$) and ($IP_{dest} \wedge M$)
2.**If** ($IP_{source} \wedge M$) = ($IP_{dest} \wedge M$) (same subnet):
(a) **If** MAC address of IP_{dest} is in ARP cache: Send packet to IP_{dest} using cached MAC
(b) **Else:**
i. Broadcast ARP request: “Who has IP_{dest} ?”
ii. Wait for ARP reply with MAC address
iii. Cache mapping $IP_{dest} \rightarrow MAC$
iv. Send packet to IP_{dest} using MAC address

DNS (domain name service). translates human-readable domain names into IP addresses.
Query initiation: Client sends DNS query to recursive resolver (typically ISP-provided)
Cache check: Resolver first checks its cache for existing resolution
Hierarchical resolution (if not cached): Resolver queries in sequence:
1.Root server (directs to appropriate TLD)
2.Top-Level Domain (TLD) server (.com)
3.Authoritative name server (holds actual domain-to-IP mapping)
Response propagation: Resolver returns IP address to client
Connection: Client uses resolved IP to connect to destination host
Distributed database: DNS mappings stored across hierarchical servers globally

Manual Testing. designed by human heuristic(intuition).
- **Exhaustive.** cover all input (not feasible)
- **Functional.** cover all requirements (depend on specification)
- **Random.** automate test generation (incomplete)
- **Structural.** cover all code (unit testing) to test everything.).
Coverage. testing metric. software flaw can only be detected if flawed statement is executed
⇒ effectiveness of a test suite depends on how many statements are executed.
- **Statement Coverage.** Measures how many statements (e.g., assignments, comparisons, etc.) in the program have been executed by the test suite.
- **Branch Coverage.** Measures how many branches among all possible execution paths have been executed by the test suite.
3. **Mutation-based Fuzzing:** uses set of valid seed inputs; modifies inputs based on feedback from previous rounds. Mutations can be informed by program structure:
White-box (full access to code/internal logic)
Grey-box (partial knowledge, typically coverage feedback only)
Black-box (no internal knowledge, input/output observation only)
Undefined Behavior Sanitizer (UBSan). (*only one usable in production.*) Detects undefined behavior by instrumenting code to trap on typical C/C++ undefined behavior: - Unsigned/misaligned pointers
- Signed integer overflow
- Conversion between floating point types leading to overflow
- Illegal use of NULL pointers
- Illegal pointer arithmetic
Performance. slowdown depends on amount/frequency of checks.

Address Sanitizer (ASan). Detects memory errors by placing red zones around objects and checking them on trigger events. It can detect:
- Out-of-bounds accesses to heap, stack, and globals
- Use-after-free
- Use-after-return (config.)
- Use-after-scope (configurable)
- Double-free and invalid free
- Memory leaks (experimental)
Performance.+ 2× slowdown.



ARP Defenses:
Static ARP entries. Manually configure critical mappings (*gateway*) in read-only cache
Spoofing detection. Monitor for inconsistencies (one IP → multiple MACs/one MAC → multiple IPs)
Cross-checking. Verify mappings with multiple hosts before trusting
Change alerts. Email admin when IP-MAC associations change, require human confirmation

Output: Packet sent to destination
3.**Else** (different subnet):
(a) **If** MAC address of $IP_{gateway}$ is in ARP cache: Send packet to gateway using cached MAC
(b) **Else:**
i. Broadcast ARP request "Who has $IP_{gateway}$?"
ii. Wait for ARP reply with gateway MAC
iii. Cache mapping $IP_{gateway} \rightarrow MAC$
iv. Send packet to gateway using MAC address
(c) Gateway forwards packet through WAN

DNS Spoofing Attacks. (*DNS lacks authentication and integrity protection.*)
Cache Poisoning. inject fake (domain, IP) pairs into resolver's cache ⇒ all clients querying resolver get malicious IP
DNS Hijacking. Intercept/modify DNS responses in transit (man-in-the-middle) → victim receives attacker-controlled IP
⇒ DoS/redirection to malicious hosts/MITM monitoring

DNS Spoofing Defenses:
DNSSEC. Authoritative servers sign DNS responses ⇒ prevents cache poisoning by verifying origin authenticity (no confidentiality)
DNS-over-HTTPS (DoH). Tunnel DNS queries over HTTPS connection → (confidentiality/integrity/authenticity)
DNS-over-TLS. Encrypt DNS traffic using TLS protocol

Border Gateway Protocol (BGP). make route-ing tables between Autonomous Systems (AS) with independent routing domains
1. Routers maintain tables of (**IP subnet—Router IP, cost**) (*no integrity/no authenticity*)
2. Routes change constantly due to faults, new contracts, infrastructure updates
3. BGP selects lowest-cost routes (real money considerations)
⚠️ Weak Authentication using short shared secrets (up to 80 ASCII bytes) and MD5-based MAC ⇒ only prevents DoS, not route integrity

Internet Protocol (IP). Enables addressing and routing of packets between hosts
No protection: No integrity/authentication for source/destination addresses, no confidentiality for data (only non-cryptographic checksum for transmission errors)
IP Spoofing Attacks.
Impersonation. Fake identity to steal resources
Man-in-the-Middle. Monitor, modify, or deny communication May require TCP hijacking for full MitM
DoS. Spoof source IP to trigger servers sending packets to victim
Virtual Private Network (VPN). Uses IPSec tunnel mode to create private network over public internet
Properties: Appears as single network, internal routing, full packet protection (confidentiality, authentication, integrity, replay)
Configuration 1: Connect two LANs securely
Configuration 2: Exit local network from different point (VPN as a Service—change apparent IP)
VPN rewrites return address to its own

TCP. Runs above IP, addresses IP limitations:
Reliability. Ensures messages received (retransmission)
Congestion control. Avoids network/host overload
Sessions. Associates messages into logical sessions
Multiplexing. Associates messages to specific applications via ports
Key fields: Ports, sequence numbers, acknowledgment numbers, window, flags (SYN, ACK, RST, FIN)

Transport Layer Security (TLS). Cryptographic protocol above TCP
Confidentiality: Symmetric encryption
Authentication: Public key cryptography (one or two-sided)
Integrity: MAC and signatures
Forward secrecy: Learning a key at one point does not reveal past sessions
State-of-the-art: TLS v1.3
Reality: Many insecure versions (TLS 1.0-1.2, SSL) still deployed due to upgrade difficulty

Denial of Service (DoS).
Prevent legitimate users from accessing a service
1. Crash victim by exploiting software flaws
2. Exhaust victim's resources (bandwidth/CPU/memory)
Teardrop Attack. Send IP packets with overlapping fragment offsets
Mechanism: Exploits bug in TCP/IP fragmentation reassembly → Packets overwrite each other ⇒ OS crashes or waits indefinitely for missing fragments
Defense: Patch OS to handle malformed fragments correctly

Firewalls. Network router connecting internal to external network; access control based on policy
Simple Packet Filter: Inspect each packet; rules based on $IP_{Src/Dest}$ /ports/protocol
⊗ Simple/fast/hardware level
⊖ Limited policies/static/header-only (no content inspection)
Limitations. Full mediation slow(read/check/write)/ can't auth principals(network level only)/ can't verify data correctness(headers lack auth)/ **Filter "bad" traffic/remove noise, NOT security solution**

De-Militarized Zone (DMZ). Defense in depth with 3 zones:
WAN: Outside world
DMZ: Public services (*email servers...*)
LAN: Internal users
Traffic rules: DMZ/LAN to Internet: **Allowed**
Internet/LAN to DMZ: **Allowed**
DMZ/Internet to LAN: **Blocked** ⇒ **If DMZ compromised, LAN remains safe**

Intrusion Detection System (IDS). monitors/alerts suspicious activity (alarm, not lock)
Detection methods: Signature-based: Matches known attack fingerprints
Anomaly-based: Flags deviations from normal behavior
Limitation: Blind to encrypted traffic without TLS inspection
Network Address Translation (NAT). Router maintains mapping: (Internal IP, port) → (External IP, port)
Security benefit: External entities can't route into NAT unless mapped port exists ⇒ reduces attack surface / *Save IPv4 address space*

BGP Hijacking Attacks. Adversary controls/ promises router, injects false low-cost routes → attracts traffic to adversary path and false routes propagate until expiration
⇒ *Surveillance, injection/modification, censorship, DoS, modifies route, not destination*

BGP Hijacking Defenses:
Route filtering. Block geographically implausible routes (⚠️ limited effectiveness)
Challenge No central authority for route correctness
⇒ **BGPsec.** AS certificates link keys to IP blocks; updates require authoritative AS signature

IPSec Defenses. uses public key/pre-shared symmetric key encryption.
Authentication Header(AH). Authentication/Integrity(via HMAC) + anti-replay protection (using sequence numbers). Protects payload (**auth only**).
Encapsulating Security Payload(ESP) Adds confidentiality by encryption (**auth+encryption**)
Transport mode. Encrypts only payload, keeps original IP headers (end-to-end)
Tunnel mode. Encrypts entire packet (headers + payload) wraps in new packet with new IP headers (gateway-to-gateway).

VPN Limitations, Does NOT protect against: DoS attacks/ Application-layer auth issues
VPN vs Proxy:
VPN: End-to-end encryption, acts as one network, no MitM
Proxy: Encryption only end-to-proxy, separates two networks, proxy sees both sides (acts as MitM)
Both hide IP from receiver but offer different security properties

TCP Hijacking Attacks. (*No integrity/authentication; seq numbers = weak secret.*)
3-way handshake:
1) Client: SYN + SeqC
2) Server: SYN-ACK + SeqC+1 + SeqS
3) Client: ACK + SeqS+1
Attack. adversary guesses sequence numbers ⇒ hijack/insert packets

TLS Handshake. Bootstrap secure communication, agree on algorithms, establish session keys
1.ClientHello: Version, CipherSuites (ordered preferences), SessionID, challenge R_C
2.ServerHello: Chosen Cipher, ServerCertificate (PKI cert with signed public key), [ServerKeyExchange] (e.g., g^a for DH), [ClientCertRequest], challenge R_S
3.Client: [ClientCertificate], ClientKeyExchange (e.g., g^b), ChangeCipherSpec flag
3.1.ClientFinish: Encrypted/authenticated "done" message
4.Server: ChangeCipherSpec flag
4.1.ServerFinish: Encrypted/authenticated "done" message
⇒ Shared session key established, all subsequent traffic encrypted & authenticated

TCP SYN Flood. Send TCP SYN packets with bogus source addresses
Mechanism: Server allocates TCP Control Block (TCB, 280 bytes) for each half-open connection and waits for ACK
→ TCB limit *exhausted* ⇒ new legitimate requests rejected
Defense: SYN cookies - compress TCP state/push state to client (cryptographically protected), require client to return it; Alternative: Proof-of-Work (require computational work before processing)
TCP RST Injection (Great Firewall of China). Inject forged TCP RST packets into established connections
Mechanism: No auth in TCP; endpoints trust RST flag and abandon connection
→ Connection terminated without flooding server (server sees client disconnect)
Why not drop packets? Dropping requires acting as MitM and physically intercepting; injection is easier
Defense: TLS/IPSec provide authentication to prevent forged packets

Stateful Firewall: Understands Application Firewall: Deep TCP/UDP semantics; decisions Packet Inspection *evaluates content*; depend on connection state *tent*; blocks P2P, scans for viruses, e.g. *allows dynamic high port* monitors leaks
connections only during active Encrypted traffic; Block all or use client certificates for decryption
FTP session
Limitations. Full mediation slow(read/check/write)/ can't auth principals(network level only)/ can't verify data correctness(headers lack auth)/ **Filter "bad" traffic/remove noise, NOT security solution**

Deployment types: NIDS: Network-based(monitors traffic)
HIDS: Host-based(monitors system)
Limitation: Blind to encrypted traffic without TLS inspection
Network Address Translation (NAT). Router maintains mapping: (Internal IP, port) → (External IP, port)
Security benefit: External entities can't route into NAT unless mapped port exists ⇒ reduces attack surface / *Save IPv4 address space*

Privacy.
1. Right to be let alone: Freedom from intrusion
2. Informational self-determination: Control over when/how/what shared
3. Freedom to construct identity: Autonomy in persona projection
Privacy IS security: Protects against profiling, manipulation, identity theft (individuals); trade secrets (companies); national secrets (governments)
Three Types of PETs.
2. Institutional
Privacy:
Adversary: Provider
Goal: GDPR compliance (consent, Social purpose limitation, data minimization, Goal: Don't surprise security, auditability)
Tech: Access control, logging, anonymization
△ Never questions collection
3. Anti-Surveillance
Privacy:
Adversary: Everyone (infrastructure)
Goal: Minimize disclosure, minimize trust
Tech: E2E encryption, anonymous comms
△ Narrow designs, usability issues

Tor Network (Onion Routing). Overlay network at application layer/low-latency anonymous communication
1. Select path: Choose 3 nodes (entry, middle, exit) from directory authorities
2. Build circuit: Authenticated DH key agreement:
- Entry: Direct key agreement
- Middle: Via entry (middle blind to user IP)
- Exit: Via entry+middle (exit blind to user IP and entry IP)
3. Send: Encrypt message in layers: $E_{k_{exit}}(E_{k_{middle}}(E_{k_{entry}}(msg)))$
Each node decrypts one layer $\Rightarrow$ bitwise unlinkability
4. Response: Exit encrypts in reverse order (inner layer for entry)
Stream-based: Circuit built once (expensive PK ops), reused with symmetric keys
Decentralized trust: Separation of privilege, no single node breaks anonymity
Assumption: Adversary cannot observe both entry and exit edges

Traffic Analysis & Metadata.
Encryption protects content, NOT metadata (IPs, timing, volume, location, device)
Traffic analysis: Extract info from communication patterns
E2E Encryption:
User devices as ends $\Rightarrow$ protects from all
Provider servers as ends (TLS) $\Rightarrow$ only from network observers
Anonymous Comms Properties:
1. Bitwise unlinkability (crypto)
2. Hide patterns (repacketize, reschedule, reroute)
Low-latency (Tor): Stream-based, vulnerable to global adversary
High-latency (Mixes): Message-based, delays, resists global adversary

Tor vs VPN.
Tor: Decentralized trust, anonymity if adversary can't see both edges
VPN: Centralized trust, no anonymity vs provider
Anonymous Credentials.
Anonymous authentication via attribute proof (not identity)
Properties:
- Prove attribute without identity (e.g., "over 18")
- Data minimization
- Users with same attribute indistinguishable
- Multiple uses unlinkable
vs PKI: Both use trusted issuer & secret; PKI reveals identity, enables tracking